

Arnav Gupta

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Summary

Software engineer with backend and distributed systems experience across cloud and financial infrastructure. Built event-driven cloud pipelines and RESTful control-plane APIs as an AWS intern, and designed distributed data processing systems (Hadoop, Spark) for high-throughput financial surveillance at Goldman Sachs. Project experience in C++ parallel programming with OpenMP, lock-free data structures, and cache-aware performance optimization.

Education

New York University – Courant

MS in Computer Science

Sept 2025 – Present

New York, USA

- GPA: 3.89/4
- Relevant Courses: Fundamental Algorithms, Numerical Methods I, Computer Graphics, Multicore Processors, Operating Systems, Programming Languages, Parallel Algorithms, GPU Architecture and Programming

Indian Institute of Technology – Delhi

B.Tech in Engineering Physics

July 2017 – May 2021

New Delhi, India

- GPA: 8.234/10
- Relevant Courses: Data Structures & Algorithms, Computational Physics, Machine Learning, Software Engineering, Probability & Stochastic Processes, Calculus, Linear Algebra & Differential Equations, Cryptography & Computer Security, Quantum Information & Computing

Experience

Software Development Engineer Intern – Amazon Connect

Amazon Web Services (AWS)

May 2026 – August 2026

New York City, USA

- Architected and built an event-driven distributed pipeline (Amazon SES to EventBridge to Kinesis to ECS) with cross-account credential management (STS) backed by a DynamoDB-based config resolution to process delivery status events asynchronously at scale, using a cellular topology with per-cell consumer isolation for fault tolerance
- Designed and built an end-to-end Email Delivery Failure Alerting system for Amazon Connect, enabling real-time agent notifications and automated task routing when outbound emails fail to deliver, across 30+ AWS regions; authored 2 technical design documents and contributed to system design decisions and code quality standards across the team
- Developed 7 RESTful control plane APIs with idempotency handling and AppConfig feature gating to allow users to configure settings for the notification and task routing system, following AWS API Design Standards with full unit and integration tests

Software Developer II (Associate)

Goldman Sachs

July 2021 – June 2025

Bangalore, India

- Part of the Financial Crime Compliance team, building robust and scalable surveillance software that handles terabytes of financial data to detect insider trading risk
- Built a high-throughput distributed surveillance pipeline processing terabytes of trading data daily, reducing alert latency on late-arriving trades and increasing detection recall by re-architecting the batch processing layer to run 3x faster
- Redesigned surveillance software for identifying manipulated penny stock products in the OTC market, migrating from single-threaded Java processes to a MapReduce-based distributed system, achieving a 6x improvement in surveillance run times
- Designed and developed surveillance software that monitors firm personnel who are 'wall-crossed' and restricted from trading on products issued by certain corporations

Software Developer Intern

Artivatic.AI

May 2020 – July 2020

Remote

- Built a multi-user video call platform using WebRTC and Django Channels, adding screenshare and call recording
- Developed an API service for insurance premium payments and payment history retrieval by integrating the Razorpay SDK
- Developed an API service for extracting information from customer legal documents (Passport, Driving License,

- ITR, and other government-issued IDs) and storing them in the database
- Optimized how legal document files were saved and served, resulting in faster dashboard loading times

Skills

Languages: Java, C++, Python, Scala, JavaScript, Bash, OCaml, Standard ML (SML/NJ)

Distributed Systems & Data Processing: Apache Spark, Apache Hadoop, MapReduce, HDFS, Apache HBase

Cloud Infrastructure (AWS): Kinesis, DynamoDB, ECS, SES, EventBridge, S3, IAM, STS, CDK, AppConfig

Backend & Real-Time: Node.js, Express.js, Django REST Framework, WebRTC, Socket.IO

Graphics: OpenGL, WebGL, GLSL

Achievements

Winner – Tech@NYU Startup Week Premier Buildathon

February 2026

2nd Place, Best AI Automation Track; 3rd Place, Best Use of Databricks

- **SRE-Agent:** An AI agent that eases the tasks of an on-call engineer – ingests production incidents, matches them to runbooks, optimizes context with Nia, uses Claude to generate a remediation plan, sends an approval request via Slack, and executes the production command via SSH/Kubernetes. Tools: Python, Databricks, Nia, Claude, Slack API, Kubernetes. Devpost

Projects

Parallel Barnes-Hut N-Body Simulation

April 2026

Tools: C++17, OpenMP – GitHub

- Implemented a 2D Barnes-Hut N-body simulator ($O(N \log N)$) in C++17 with OpenMP, scaling to 1M particles on a 64-core NUMA system (4 AMD Opteron 6272 processors)
- Designed lock-free, task-based parallel tree construction using `std::atomic` pool allocation, combined with Morton key (Z-order curve) spatial sorting to improve cache locality during force computation
- Morton sorting alone drove a 3.64x speedup in force computation, cutting the LLC miss rate 20x (20.53% to 1.03%) and improving IPC 3.71x (0.14 to 0.52), measured via hardware performance counters (`perf stat`)
- Measured a 36x speedup at 64 threads for 100K particles; evaluated scalability with strong-scaling experiments across problem sizes (1K to 1M particles) and accuracy settings (theta values from 0.2 to 1.0)

Acadmaze – Back-End Developer Intern

March 2019 – June 2019

Node.js, Express.js, Socket.IO, MongoDB, Amazon S3

- Web-based social network for university students, professors, and industry to collaborate on research and academic projects
- Built a secure login interface with session management, forgot-password, and email verification
- Built a real-time notification system for events, posts, and messages, and made post/comment creation asynchronous
- Integrated Amazon S3 for storing user interactions, posts, messages, and media
- Built a pagination system for the homepage feed to improve load time

Transient Plasma Photonic Crystal (TPPC) Simulations

August 2020 – November 2020

Tools: EPOCH

- Studied formation of Transient Plasma Photonic Crystals from underdense plasma using counter-propagating lasers (G. Lehmann and K. H. Spatschek, Phys. Rev. E 97, 063201 (2018))
- Conducted 1D and 2D particle-in-cell simulations with the EPOCH PIC code, demonstrating TPPC formation and its use as a reflector and polarizer

CURE Algorithm Implementation

Feb 2020 – March 2020

Tools: Python – GitHub

- Implemented the CURE clustering algorithm (more robust to outliers and non-spherical clusters than K-means) using K-D Trees and heaps
- Visualized clustering results on the Old Faithful geyser dataset